🩺 UAE Cancer Data Analysis Report

🧭 Objective

This project explores UAE cancer patient data to:

- Understand patient outcomes across demographic and lifestyle factors.
- Identify patterns by cancer types, smoking status, and comorbidities.
- Identify the top five most common cancer types.
- Build predictive models (XGBoost) to forecast outcomes.
- Use clustering (KMeans) for profiling patient subgroups.

📌 **Dataset Source:** Provided by **Internpulse**
📁 **Filename:** _cancer_dataset_uae.xlsx

🛠️ Libraries Used

python

import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.model_selection import train_test_spli

from sklearn.ensemble import RandomForestClassifier

from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score, classification_report, confusion_matrix

from sklearn.decomposition import PCA

import xgboost as xgb

🔍 Methodology

1. **Data Cleaning**
   - Removed duplicates and missing values.
   - Converted Diagnosis_Date to datetime, extracted Diagnosis_Year.
   - Encoded categorical variables using LabelEncoder.
2. **Exploratory Data Analysis (EDA)**
   - Countplots, histograms, and line charts were used to visualize outcomes.
   - Grouped data by Cancer_Type, Smoking_Status, Gender, and more.
3. **Modeling**
   - XGBoost classifier was used for predicting patient outcomes.
   - Model accuracy was displayed and used for performance evaluation.
4. **Unsupervised Learning**
   - KMeans clustering used to identify patterns in patient profiles.
   - Optimal clusters chosen using Elbow Method and Silhouette Score.

📈 Key Insights with Figures

1. Patient Outcome Distribution

- **Recovered:** ~2,356 patients
- **Deceased:** ~356 patients
- Recovered patients outnumber deceased significantly, indicating effective treatment strategies for certain cancer types.

2. Recovery Rate by Cancer Type (Top Performers)

Cancer Type	Recovery Rate
Skin Cancer	0.93

Cancer Type	Recovery Rate
Prostate Cancer	0.89
Breast Cancer	0.86
Colon Cancer	0.84
Lymphoma	0.81

🔎 **Insight:** Skin and prostate cancer types have the highest recovery rates, suggesting either early detection or more effective treatments for these categories.

3. Smoking Status vs. Outcome

- **Smokers** had a **significantly higher proportion of deaths** compared to non-smokers.
- In visual countplots, the bar for "Deceased" was much higher for smokers.

💡 **Insight:** Smoking status is a strong negative predictor of survival, highlighting the importance of anti-smoking interventions.

4. Gender and Ethnicity Distributions

- **Male Patients:** ~58%
- **Female Patients:** ~42%
- **Most Common Ethnicities:** Emirati, South Asian, and Arab

📌 **Insight:** A slightly higher prevalence among males may inform gender-focused health campaigns. South Asians and Emiratis formed the bulk of cases, suggesting localized public health focus.

5. Top 5 Most Diagnosed Cancer Types

Cancer Type	Frequency
Breast Cancer	280
Lung Cancer	230
Leukemia	190
Prostate Cancer	160

Cancer Type	Frequency
Colon Cancer	150

💬 **Insight:** These top cancer types should be prioritized for resource allocation, awareness, and screening programs.

6. Model Performance (XGBoost Classifier)

- **Accuracy:** ~85%
- **Top Features:** Smoking Status, Age, Comorbidities

📊 **Insight:** Machine learning can reliably predict patient outcomes. These features can be used in risk calculators or early warning systems.

7. Clustering (KMeans)

- Optimal clusters found: **3**
- Grouped patients based on Age, Height, Weight, and Lifestyle factors.
- One cluster had predominantly younger, non-smoking patients with high survival rates.

🧬 **Insight:** Unsupervised learning can segment patients into low-risk and high-risk groups, aiding personalized treatment.

💡 Recommendations

1. **Implement Nationwide Anti-Smoking Campaigns**
   Strong link between smoking and mortality should be a public health priority.
2. **Prioritize Screening for Top 5 Cancers**
   Early detection in Breast, Lung, and Prostate cancers can drastically improve outcomes.
3. **Adopt Predictive Tools in Hospitals**
   XGBoost-based models can flag high-risk patients during initial diagnosis.
4. **Target Resources to Low-Recovery Cancers**
   Cancers like Lung and Pancreatic with low recovery rates need research investment.

5. **Use Patient Clustering for Personalization**
   Leverage unsupervised learning models to recommend tailored treatment plans.

🧩 Challenges Faced

Challenge	**Description**	**Solution**
Missing values	Found in Diagnosis_Date, Ethnicity, and Smoking_Status	Used imputation and dropped invalid rows
Imbalanced class (Outcome)	‘Recovered’ cases outnumbered ‘Deceased’	Stratified sampling and XGBoost handling
Clustering sensitivity	Hard to determine k manually	Used Elbow Method and Silhouette Score
Text overlap in visualizations	Label clutter in plots	Added annotations and rotated x-ticks

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